GUI Testing of "HomeResq"

A MINI PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT FOR THE AWARD OF THE DEGREE OF MASTER IN COMPUTER APPLICATIONS

Submitted by

Manushree Ray(25MCA20242)

Supervisor

Ms. Priyanka_E18375

ASSISTANT PROFESSOR, UIC CHANDIGARH UNIVERSITY

UNIVERSITY INSTITUTE OF COMPUTING
CHANDIGARH UNIVERSITY
MOHALI, PUNJAB-140301

ACKNOWLEDGEM EMNT

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Manushree Ray

25MCA20242

Abstract

HomeResQ is an advanced web-based application designed to provide immediate and efficient solutions for home-related emergencies such as plumbing issues, electrical failures, medical assistance, and security concerns. In today's fast-paced lifestyle, individuals often face difficulties in finding reliable service providers during urgent situations. This project aims to bridge that gap by offering a centralized and user-friendly platform where users can quickly request emergency services and receive timely assistance.

The system is developed using modern web technologies to ensure high performance, responsiveness, and scalability. The frontend of the application is built using HTML, CSS, and JavaScript, which provide a structured layout, visually appealing design, and dynamic user interaction. These technologies work together to create an intuitive and responsive interface that can be accessed across multiple devices, including desktops, tablets, and smartphones.

On the backend, Node.js is used as the runtime environment to handle server-side operations, manage API requests, and ensure smooth communication between the user interface and the database. It enables efficient handling of multiple user requests simultaneously, making the system fast and reliable. The application uses MongoDB as its database, which is a NoSQL database known for its flexibility, scalability, and ability to handle large volumes of data. MongoDB stores important information such as user profiles, service requests, service provider details, and transaction records in a structured and secure manner.

Additionally, the system includes features such as user authentication, emergency request handling, and an admin panel for managing and monitoring service requests. Overall, HomeResQ provides a reliable and efficient platform that improves response time, enhances accessibility, and ensures better user experience in emergency situations.

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Introduction:

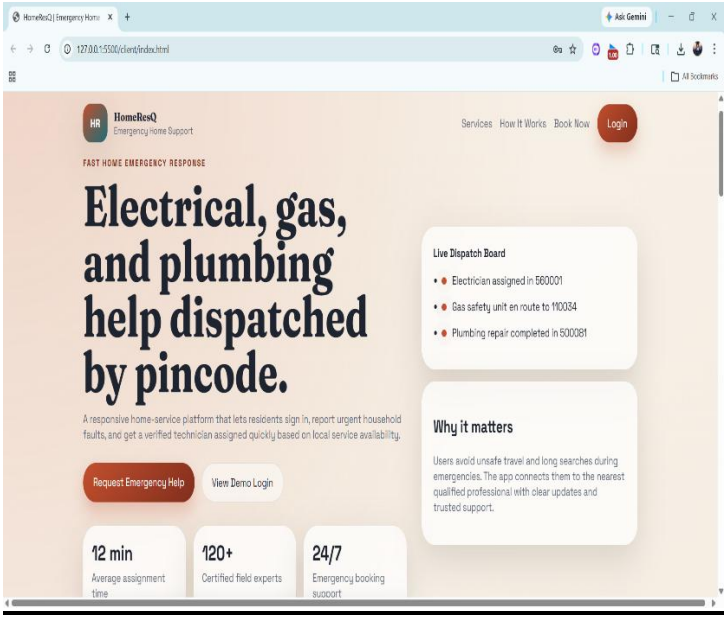
Graphical User Interface (GUI) Testing is an essential part of software testing that focuses on verifying the visual elements and user interaction of an application. It ensures that all interface components such as buttons, forms, menus, links, images, icons, and layouts are functioning correctly and providing a consistent, smooth, and user-friendly experience. GUI testing not only checks the appearance of the system but also validates how users interact with the application in real-time scenarios.

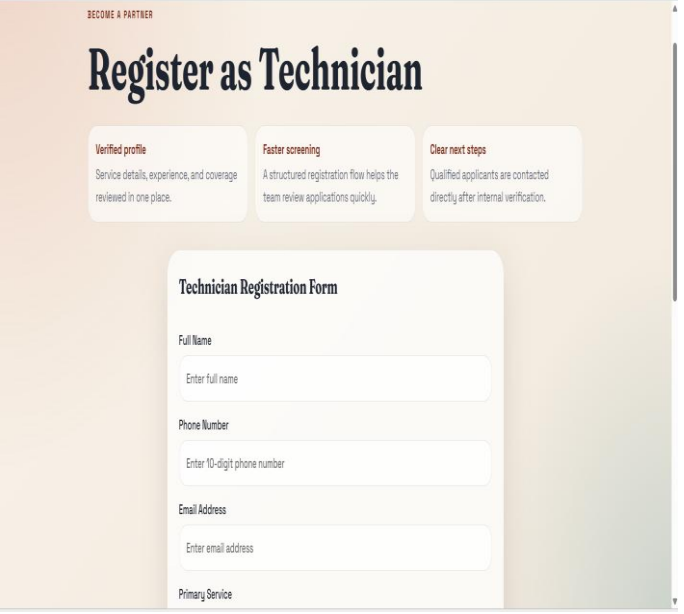
In the context of the **Home ResQ** system, GUI testing plays a highly critical role because the application is designed to provide emergency services where time and accuracy are extremely important. Users rely on the platform during urgent situations such as plumbing failures, electrical issues, medical emergencies, or security threats. In such cases, even a minor issue in the interface, such as a non-working button, incorrect navigation, or missing validation message, can delay service requests and negatively affect the user experience. Therefore, it is necessary to ensure that the interface is simple, responsive, and error-free.

Objective:

The main objective of GUI Testing in the Home ResQ system is to ensure that the user interface is visually consistent, functionally correct, and easy to use for all users, especially during emergency situations where quick interaction is critical. The specific objectives are as follows:

- To verify proper display and functionality of UI elements such as buttons, forms, images, and links
- To validate input fields and ensure correct form submission with proper error handling
- To ensure smooth and correct navigation between different pages of the application
- To maintain consistency in layout, fonts, colors, and overall design





Needs :

The GUI testing of the HomeResQ system is essential to ensure that the user interface is reliable, user-friendly, and efficient, especially during emergency situations where quick interaction is required. Since the application deals with urgent service requests, any issue in the interface such as improper layout, non-functional buttons, or missing validations can lead to delays and inconvenience for users.

The needs for GUI testing are summarized as follows:

- To ensure that all user interface elements (buttons, forms, menus, and links) are properly displayed and aligned (**GUI Testing**).
- To verify that all buttons and interactive components function correctly without errors (**Functional Testing**).
- To validate input fields and ensure proper error messages are shown for invalid data (**Validation Testing**).
- To ensure smooth and correct navigation between different pages of the application (**Usability Testing**).
- To maintain consistency in design, including fonts, colors, and layout (**GUI Testing**).
- To check responsiveness of the interface across different devices such as mobile, tablet, and desktop (**Responsiveness Testing**).

Functional Requirements:

Functional requirements define what the system should do and the features it must provide.

1. **User Module**
   - User registration with required details
   - User login and authentication
   - View available emergency services
   - Request emergency service (plumbing, electrical, medical, etc.)
   - View request status
2. **Service Module**
   - Display list of services
   - Allow users to select and request a service
   - Store service request details
3. **Admin Module**
   - Admin login
   - View all service requests
   - Approve or reject requests
   - Manage users and services.
4. **GUI Functional Requirements**
   - Buttons (Login, Register, Request Service) must be clickable
   - Forms must accept valid input and reject invalid input
   - Navigation links must redirect correctly

Non-Functional Requirements:

1. **Performance**
   - Fast response time for service requests
   - Ability to handle multiple users simultaneously
2. **Usability**
   - Simple and user-friendly interface
   - Easy navigation
   - Clear instructions and layout
3. **Responsiveness**
   - Works on mobile, tablet, and desktop
   - Layout adjusts to different screen sizes
4. **Security**
   - Secure login authentication
   - Protection of user data
   - Prevention of unauthorized access

Features:

The **Home ResQ system** provides several important features to ensure quick, reliable, and user-friendly emergency service management. The key features are as follows:

1. **User Registration and Login**
   - Allows users to create an account
   - Secure login authentication
   - Easy access to services
2. **Emergency Service Request**
   - Users can request services like:
     - Plumbing
     - Electrical
     - Medical assistance
     - Security services
   - Quick and simple request process
3. **Centralized Platform**
   - All services available in one place
   - Eliminates need to search manually
4. **Admin Dashboard**
   - Admin can:
     - View service requests
     - Approve or reject requests
     - Manage users and services
5. **Real-Time Request Handling**
   - Immediate request submission
   - Faster response to emergencies

Methodology:

The development and testing of the HomeResQ system follow a structured approach to ensure a reliable and user-friendly application.

- **Requirement Analysis** – Identify system needs like login, registration, and service requests
- **System Design** – Plan architecture, UI layout, and data flow
- **Development** – Build frontend (HTML, CSS, JS) and backend (Node.js)
- **Database Design** – Use MongoDB to store user and service data
- **API & CRUD** – Connect frontend-backend and manage data operations
- **Testing** – Perform GUI and functional testing to ensure proper working

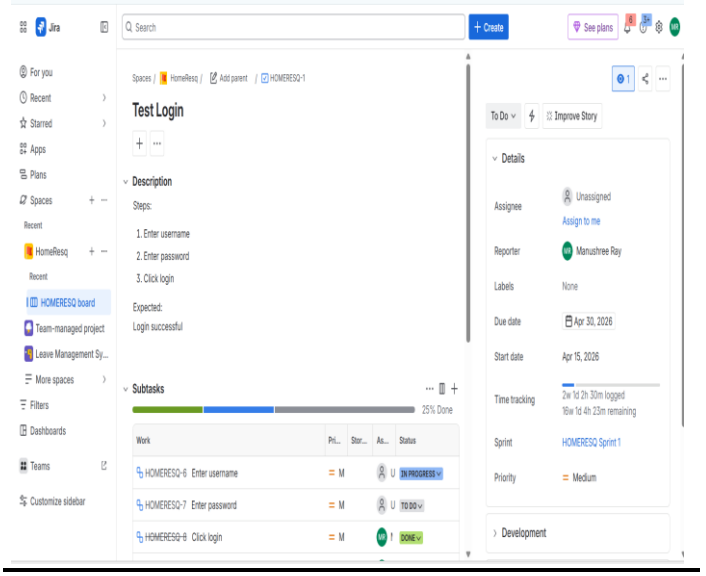
Future Work:

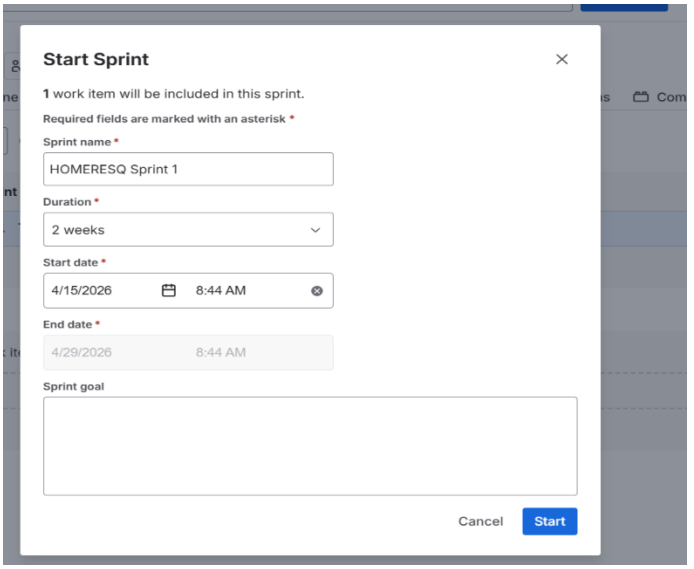
The **HomeResQ system** can be further improved and enhanced with additional features to make it more efficient, reliable, and user-friendly. Future developments can expand the system's capabilities and improve overall performance.

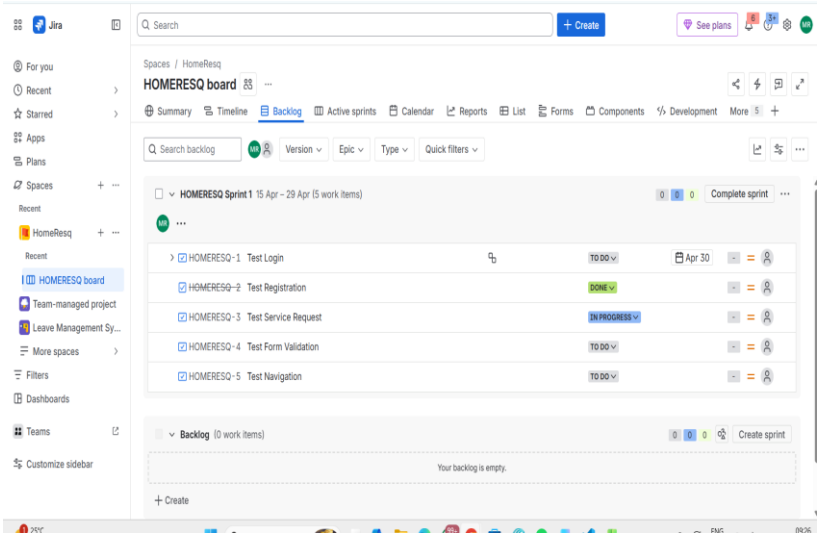
Future enhancements may include:

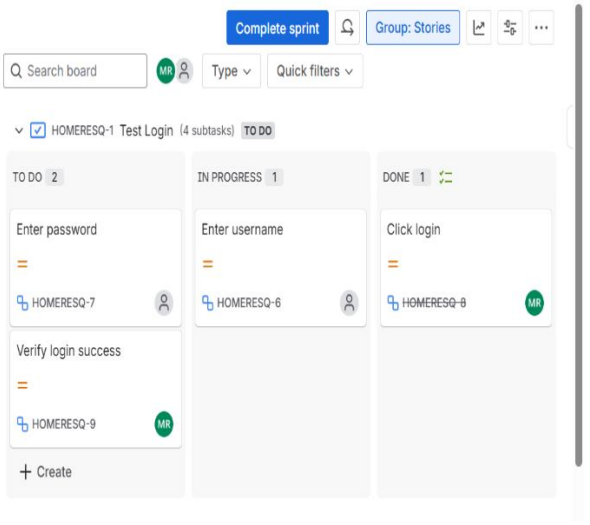
1. **Real-Time Location Tracking** – Integrate GPS functionality to track service providers and provide accurate arrival time.
2. **Online Payment Integration** – Add secure payment options for seamless transactions within the platform.
3. **User Authentication and Security** – Implement advanced security features such as OTP verification and role-based access.
4. **Mobile Application Development** – Develop Android and iOS applications for better accessibility and convenience.
5. **AI-Based Service Recommendation** – Use artificial intelligence to suggest services based on user needs and past activities.
6. **Live Chat Support** – Provide real-time communication between users and service providers.
7. **Cloud Integration** – Store data on cloud platforms for better scalability, backup, and remote access.

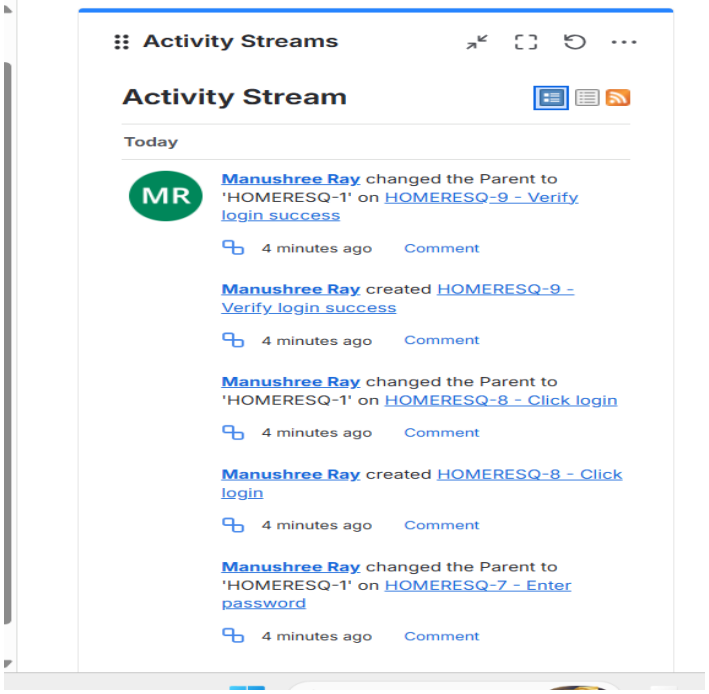
Output:

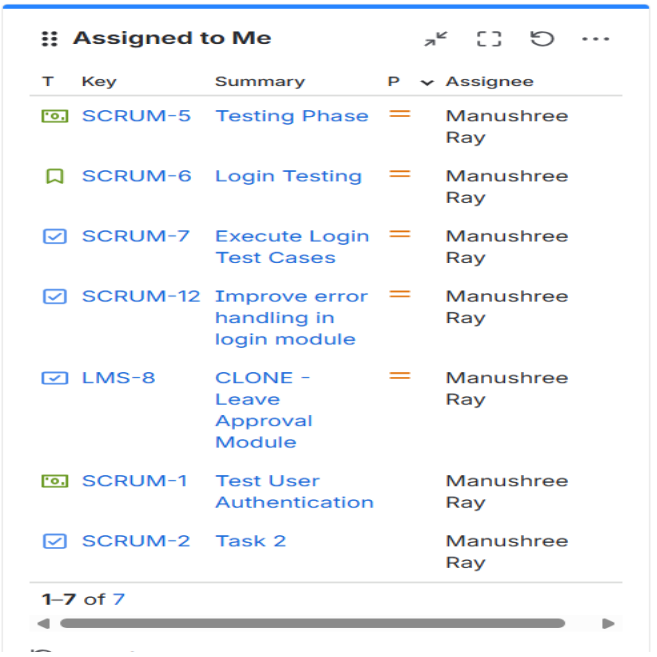

Test Login
Description
Steps:
1. Enter username
2. Enter password
3. Click login
Expected:
Login successful
Subtasks
Details
Assignee
Reporter
Labels
Due date
Start date
Time tracking
Sprint
Priority
Development

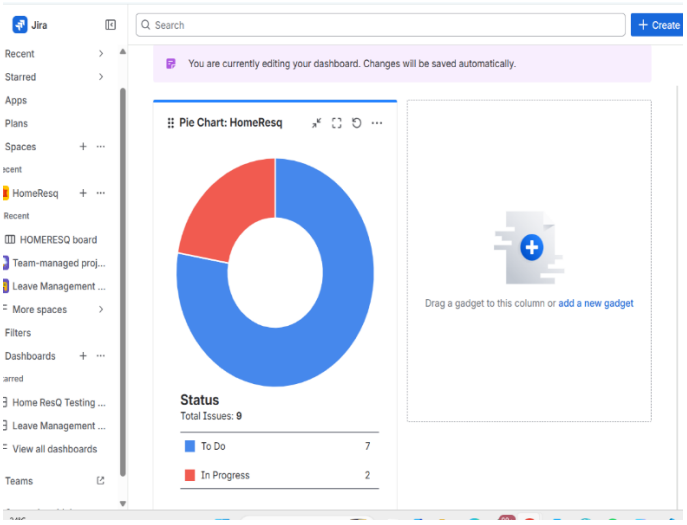

Start Sprint
1 work item will be included in this sprint.
Required fields are marked with an asterisk *
Sprint name *
HOMERESQ Sprint 1
Duration *
2 weeks
Start date *
4/15/2026
8:44 AM
End date *
4/29/2026
8:44 AM
Sprint goal
Cancel
Start

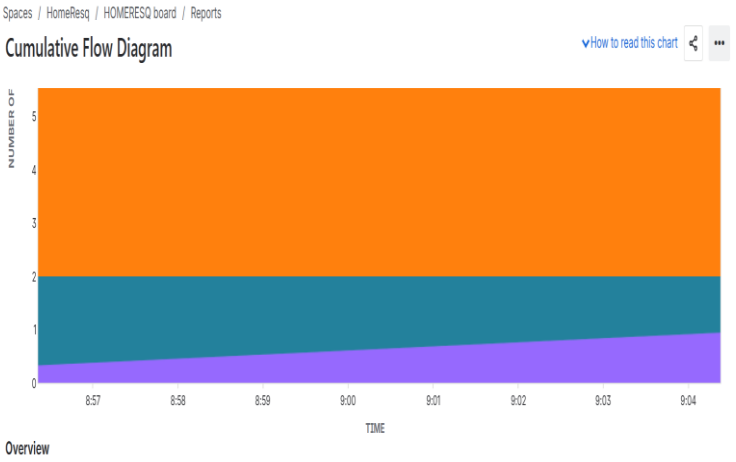

HOMERESQ board
Summary
Timeline
Backlog
Active sprints
Calendar
Reports
List
Forms
Components
Development
HOMERESQ Sprint 1 15 Apr – 29 Apr (5 work items)
Complete sprint
HOMERESQ-1 Test Login
HOMERESQ-2 Test Registration
HOMERESQ-3 Test Service Request
HOMERESQ-4 Test Form Validation
HOMERESQ-5 Test Navigation
Backlog (0 work items)
Create sprint
Your backlog is empty.


Complete sprint
Group: Stories
Search board
Type
Quick filters
HOMERESQ-1 Test Login (4 subtasks) TO DO
TO DO 2
IN PROGRESS 1
DONE 1
Enter password
HOMERESQ-7
Enter username
HOMERESQ-6
Click login
HOMERESQ-8
Verify login success
HOMERESQ-9
Create


Activity Streams
Activity Stream
Today
MR
Manushree Ray changed the Parent to 'HOMERESQ-1' on HOMERESQ-9 - Verify login success
4 minutes ago Comment
Manushree Ray created HOMERESQ-9 - Verify login success
4 minutes ago Comment
Manushree Ray changed the Parent to 'HOMERESQ-1' on HOMERESQ-8 - Click login
4 minutes ago Comment
Manushree Ray created HOMERESQ-8 - Click login
4 minutes ago Comment
Manushree Ray changed the Parent to 'HOMERESQ-1' on HOMERESQ-7 - Enter password
4 minutes ago Comment


Assigned to Me
T Key Summary P Assignee
SCRUM-5 Testing Phase Manushree Ray
SCRUM-6 Login Testing Manushree Ray
SCRUM-7 Execute Login Test Cases Manushree Ray
SCRUM-12 Improve error handling in login module Manushree Ray
LMS-8 CLONE - Leave Approval Module Manushree Ray
SCRUM-1 Test User Authentication Manushree Ray
SCRUM-2 Task 2 Manushree Ray
1–7 of 7





Module - Test Cases – HomeResQ

Module-wise testing is performed to verify each part (module) of the **HomeResQ system** individually. This helps in identifying errors in specific functionalities and ensures that all modules work correctly before integration.

1. User Module Test Cases

The User Module handles user-related operations such as registration, login, and authentication.

Testing Activities:

- Testing user registration with valid and invalid inputs
- Verifying login functionality with correct and incorrect credentials
- Checking error messages for invalid login attempts
- Testing logout functionality

TC ID	Test Case	Steps	Expected Result	Actual Result	Status
UM-01	User Registration	Enter valid details & submit	Registration successful	Registration successful	Pass
UM-02	Invalid Registration	Enter invalid email	Error message shown	Error message displayed	Pass
UM-03	Login with valid data	Enter correct username & password	Login successful	Login successful	Pass
UM-04	Login with invalid data	Enter wrong credentials	Error message shown	Error message displayed	Pass
UM-05	Logout	Click logout button	User logged out	User logged out successfully	Pass

2. Service Request Module Test Cases

This module allows users to select emergency services and submit service requests.

Testing Activities:

- Selecting different service categories (plumbing, electrical, medical, etc.)
- Filling and submitting service request forms
- Verifying successful request submission
- Checking system response for incomplete forms
-

TC ID	Test Case	Steps	Expected Result	Actual Result	Status
SR-01	Select Service	Choose service option	Service selected	Service selected successfully	Pass
SR-02	Submit Request	Fill form & submit	Request submitted	Request submitted successfully	Pass
SR-03	Empty Form	Submit without data	Error message shown	Form submitted without error	Fail
SR-04	Confirmation Message	Submit valid request	Confirmation displayed	Confirmation message displayed	Pass

3. Form Validation Module Testing

This module ensures that user inputs are valid and complete before submission.

Testing Activities:

- Submitting forms with empty fields
- Entering invalid data (wrong email, phone number, etc.)
- Checking validation messages

TC ID	Test Case	Steps	Expected Result	Actual Result	Status
FV-01	Empty Fields	Leave fields empty	Validation error	Validation error displayed	Pass
FV-02	Invalid Email	Enter wrong email	Error message	Error message displayed	Pass
FV-03	Invalid Phone	Enter wrong number	Error message	Error message displayed	Pass

4. Admin Module Test Cases

The Admin Module allows administrators to manage user requests and system operations.

Testing Activities:

- Viewing all service requests
- Approving or rejecting requests
- Updating request status

TC ID	Test Case	Steps	Expected Result	Actual Result	Status
AD-01	View Requests	Open admin panel	Requests displayed	Requests displayed correctly	Pass
AD-02	Approve Request	Click approve	Status updated	Status updated successfully	Pass
AD-03	Reject Request	Click reject	Status updated	Status updated successfully	Pass

Boundary Value Analysis in HomeResQ

Boundary Value Analysis (BVA) is a testing technique used to verify the behavior of the system at the **boundary limits (minimum and maximum values)** of input fields. In the **HomeResQ system**, BVA is applied to ensure that the application correctly accepts valid inputs within the defined range and rejects invalid inputs outside the range.

Username Length:

Test Case	Input Value	Expected Result	Actual Result	Reason	Status
TC-01	2 characters	Reject	Rejected	Less than minimum limit	Pass
TC-02	3 characters	Accept	Accepted	Minimum valid value	Pass
TC-03	10 characters	Accept	Accepted	Maximum valid value	Pass
TC-04	11 characters	Reject	Rejected	Greater than maximum limit	Pass

Phone Number:

Test Case	Input Value	Expected Result	Actual Result	Reason	Status
TC-05	9 digits	Reject	Rejected	Less than required digits	Pass
TC-06	10 digits	Accept	Accepted	Valid phone number	Pass
TC-07	11 digits	Reject	Rejected	More than required digits	Pass

Decision Table:

A **Decision Table** is a testing technique used to represent different conditions and their corresponding actions in a tabular form. It helps in checking all possible combinations of inputs and ensures that the system behaves correctly in each case.

Decision Table – Login Module

Conditions / Rules	Rule 1	Rule 2	Rule 3	Rule 4
Username entered	Yes	Yes	No	No
Password entered	Yes	No	Yes	No
Action				
Login Success	Yes	No	No	No
Show Error	No	Yes	Yes	Yes

Decision Table – Service Request

Conditions / Rules	Rule 1	Rule 2	Rule 3
Service selected	Yes	No	Yes
Form filled	Yes	Yes	No
Action			
Request Submit	Yes	No	No
Show Error	No	Yes	Yes

Conclusion:

The GUI testing of the **HomeResQ system** plays a vital role in ensuring that the application delivers a smooth, efficient, and user-friendly interface. Since the system is designed to handle emergency service requests, it is extremely important that users can interact with the application quickly and without any confusion. Any delay or error in the interface can directly impact the user experience, especially during urgent situations.

Through the GUI testing process, all major interface components such as buttons, forms, navigation menus, images, and layout structures were thoroughly examined. Each element was tested to verify proper display, alignment, and functionality. The testing process helped in identifying various issues related to form validation, incorrect input handling, UI alignment, and user interaction. These issues were carefully analyzed and resolved, which significantly improved the overall usability and reliability of the system.

Additionally, the testing ensured that the application maintains consistency in design, including fonts, colors, and layout across all pages. This consistency enhances the visual appeal and makes the application easier to use. Special attention was also given to responsiveness testing, where the system was tested on different devices such as desktops, tablets, and mobile phones. The results confirmed that the application adapts well to different screen sizes and provides a seamless user experience on all platforms.